ASSIGNMENT

1 The last digit of the square of a natural number is 6. Prove that its next to last digit is odd.

Solution:

Since the last digit of its square is 6, the given natural number was even. The square of an even number is divisible by 4. Hence, the number formed by its two last digits must be divisible by 4. It is easy to write all two digit numbers which end with 6 and are multiples of 4:16, 36, 56, 76, 96. All their tens digits are odd

2 The next to last digit of the square of a natural number is odd. Prove that its last digit is 6

Solution:

For $n > 3^2$, $n^2 = (10a + b)^2 = 100a^2 + 20ab + b^2$. The number $100a^2 + 20\,ab$ has units digit 0 and an even tens digit. If the tens digit carried from b^2 is odd, then $b = 4$ or 6 only, so $b^2 = 16$ or 36, i.e. the unit digit of n^2 must be 6.

3 Is it possible to write a perfect square using only the digits (a) 2,3,6; (b) 1,2,3 exactly 10 times each?

Solution:

Every square of the form 9k, 9k+1, 9k+4 or 9k+7

(a) $x^2 \equiv 10(2+3+6) \equiv 2 \bmod (9)$

Not possible for square

(b) $x^2 \equiv 10(1+2+3) \equiv 6 \bmod (9)$

Not possible for square

4 Can the sum of the digit of a perfect square be equal to 1970?

Solution:

$x^2 \equiv 1970 \equiv 8 \bmod (9)$

Not possible for square

5 Is there a three digit number $\overline{abc}$ $(\text{where}\, a \neq c)$ such that $\overline{abc} - \overline{cba}$ is a perfect square?

Solution:

$\overline{abc} - \overline{cba} = 99a - 99s = 99(a-c) = 9 \times 11(a-c)$

For perfect square a – c must be multiple of 11 which is not possible.

6 Determine if there is a natural number k such that the sum of the two numbers $3k^2+3k-4$ and $7k^2-3k+1$ is a perfect square.

Solution:

$3k^2 + 3k - 4 + 7k^2 - 3k + 1$

$= 10k^2 - 3 \equiv 2 \bmod (5)$

Not possible for square

OR $10k^2 - 3$ has unit digit 7, not possible for square

7 If (x – 1) (x + 3) (x–4) (x – 8) + m is a perfect square, then m is

Option:

(a) 32 (b) 24 (c) 98 (d) 196

Answer: (d)

Solution:

$(x-1)(x+3)(x-4)(x-8) + m$

$(x-1)(x-4)(x+3)(x-8) + m$

$(x^2 - 5x + 4)(x^2 - 5x - 24) + m$

$x^2 - 5x + 4 = t$

$t(t-28) + m$

$t^2 - 28t + m$

$(t-14)^2 + m - 196$

For m = 196, it's perfect square

8 If n + 20 and n – 21 are both perfect squares, where n is a natural number, find n.

Solution:

$n + 20 = b^2$

$n - 21 = c^2$

$41 = b^2 - c^2 = (b-c)(b+c)$

$b + c = 41$ & $b - c = 1$

$b = 21$

$n = 421$

9 Find the maximum integer x such that $4^{27} + 4^{10000} + 4^x$ is a perfect square

Solution:

Since $4^{27} + 4^{10000} + 4^x = 2^{54} + 2^{20000} + 2^{2x} = 2^{54}(1+2.2^{1945} + 2^{2x-54})$, it is obvious that the right hand is a perfect square if $2^{2x-54} = (2^{1945})^2$, i.e., $x - 27 = 1945$, $x = 1972$

When $x > 1972$, then

$(2^{x-27})^2 = 2^{2x-54} < 1+2.2^{1945} + 2^{2x-54} < (2^{x-27}+1)^2$, So $1 + 2.2^{1945} + 2^{2x-54}$ is not a perfect square. Thus the maximal required value of x is 1972.

10 Prove that for any positive integer n,$n^4 + 2n^3 + 2n^2 + 2n + 1$ is not a perfect square

Solution:

$n^4 + 2n^3 + n^2 + 2n + 1$

$n^2 (n + 1)^2 + (n + 1)^2 = (n + 1)^2 (n^2 + 1)$

let $n^2 + 1 = p^2$

$p^2 - n^2 = 1$

$p + n = 1, p - n = 1$

$p = 1, n = 0$ (contradiction)

OR

$(n^2 + n)^2 < n^4 + 2n^3 + 2n^2 + 2n + 1 < n^4 + 2n^3 + 3n^2 + 2n + 1 = (n^2 + n + 1)^2$

11 If a nine digit number is formed by the nine non-zero digits, and its unit digit is 5, prove that it must not be perfect square.

Solution:

We prove by contradiction. Suppose that the $D = n^2$ satisfies all the requirements.

Theunit's digits of D is 5 implies that n is too.

Assume $n = 10a + 5$, then $D = (10a + 5)^2 = 100a (a + 1) + 25$, so the last two digits of D are 25.

Since the last digit of a (a+1) is 0,2 or 6 and 0,2 are impossible,

so the third digit of D is 6, ie. $D = 1000b + 625$ for some digit b.

Thus, $5^3 | D$, hence $5^4 | D$ since it's a perfect square.

However, it implies that $5^4 | 1000K$, so $5| k$. i.e. k = 0 or 5, a contradiction

12 Prove that there is no three digit number $\overline{abc}$, such that $\overline{abc} + \overline{bca} + \overline{cab}$ is a perfect square.

Solution:

$\overline{abc} + \overline{bca} + \overline{cab}$

$= 111 (a + b + c)$

$= 3 \times 37 (a + b + c)$

a + b + c must contain 3 & 37

but $a + b + c \leq 27$

$\Rightarrow$ Not possible

13 Prove that the equation $a^2 + b^2 - 8c = 6$ has no integer solution

Solution:

$a^2 + b^2 - 8c = 6$

$a^2 + b^2 = 8c + 6$

$a^2 \equiv 0,1,4 \bmod (8)$

$b^2 \equiv 0,1,4 \bmod (8)$

$a^2 + b^2 \equiv 0,1,4,2,5 \bmod (8)$

$\Rightarrow a^2 + b^2 \neq 8c + 6$

14 Show that if x and y are positive integers such that $x^2 + y^2 - x$ is divisible by 2xy, then x is a perfect square.

Solution:

From assumption, there exists integer k such that $x^2 + y^2 - x = 2kxy$.

Consider the quadratic equation in y

$y^2 - 2kxy + (x^2 - x) = 0$

Since it has an integer solution , so, by Viete Theorem, the other root is also an integer and the Discriminant Δ of the equation is perfect square. Hence

$\Delta = 4\left[k^2x^2 - \left(x^2 - x\right)\right] = 4x\left[\left(k^2 - 1\right)x + 1\right]$ is a perfect. Since x and $(k^z - 1)$ x + 1 are relatively prime, x and $(k^2 - 1)$ x + 1 both are perfect square numbers, so x is perfect square.